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Therapeutic donor hypothermia following brain death to improve the quality of transplanted organs (Review)

Amarnath DR, Tingle SJ, Hoather TJ, Thompson ER, Wilson CH, supported by Cochrane Kidney and Transplant

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[Intervention Review]

Therapeutic donor hypothermia following brain death to improve the quality of transplanted organs

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ABSTRACT

Rationale

Solid organ transplantation is the optimal treatment for patients living with end-stage organ failure. Globally, the majority of organ donors are deceased, most commonly following brainstem death. The process of brain death is known to cause significant injury to organs. Therefore, efforts to optimise the pre-donation care of these donors should be maximised to achieve improved recipient outcomes.

Objectives

To assess the benefits and harms of therapeutic donor hypothermia in recipients of organs donated from brain-dead donors.

Search methods

We searched Cochrane Kidney and Transplant's Specialised Register, CENTRAL, MEDLINE, Embase and two trials registers up to 03 September 2025.

Eligibility criteria

All randomised controlled trials (RCTs) and quasi-RCTs (RCTs in which allocation to treatment was obtained by alternation, use of alternate medical records, date of birth or other predictable methods) investigating therapeutic donor hypothermia compared with donor normothermia following brainstem death to improve the quality of transplanted organs.

Outcomes

Critical outcomes were post-transplant outcomes (delayed graft function (DGF), graft survival), utilisation and donor adverse events. Important outcomes were patient survival, primary nonfunction, acute rejection, quality of life and cardiovascular disease.

Risk of bias

We used Cochrane's risk of bias 1 tool.

Synthesis methods

Two review authors independently selected trials for inclusion, assessed methodological quality and risk of bias, and extracted data. We used random-effects models for the meta-analysis. We assessed the certainty of evidence using the GRADE approach.

Included studies

We included four studies, enrolling 2096 donors. Two studies compared donor hypothermia versus normothermia in kidney transplantation. Two studies investigated kidney, heart, lung, liver and pancreas transplantations.

Synthesis of results

Four studies assessed the impact of donor hypothermia compared with normothermia on kidney outcomes. Donor hypothermia may not reduce the rate of kidney DGF compared to donor normothermia (RR 0.87, 95% CI 0.71 to 1.08; I^2 = 57%; 4 studies, 3015 participants; low-certainty evidence).

Donor hypothermia may not be associated with one-year kidney graft survival (HR 0.77, 95% CI 0.51 to 1.14; I^2 = 0%; 3 studies, 2075 participants; low-certainty evidence) or one-year patient survival in kidney transplant recipients (HR 0.81, 95% CI 0.42 to 1.57; 1 study, 526 participants; low-certainty evidence). No studies assessed the impact on primary non-function of the kidney or acute rejection.

Evidence from only one RCT was available for other solid organ post-transplant outcomes. Donor hypothermia may show no difference to graft survival (lung transplant: HR 0.73, 95% CI 0.21 to 2.51; 1 study; low-certainty evidence; liver transplant: HR 0.85, 95% CI 0.35 to 2.06; 1 study; low-certainty evidence).

Donor hypothermia has no impact on liver utilisation (RR 1.02, 95% CI 0.96 to 1.08; I^2 = 0%; 2 studies, 1259 participants; high-certainty evidence) and probably has no impact on the utilisation of the following organs: kidney (RR 0.99, 95% CI 0.95 to 1.03; I^2 = 44%; 4 studies, 4265 participants; moderate-certainty evidence); heart (RR 1.03, 95% CI 0.87 to 1.21; I^2 = 0%; 2 studies, 1259 participants; moderate-certainty evidence); lung (RR 0.97, 95% CI 0.85 to 1.11; I^2 = 0%; 2 studies, 2518 participants; moderate-certainty evidence); pancreas (RR 1.09, 95% CI 0.74 to 1.61; I^2 = 0%; 2 studies, 1259 participants; moderate-certainty evidence). Donor hypothermia may not affect the occurrence of donor adverse events (RR 1.28, 95% CI 0.45 to 3.62; I^2 = 12%; 3 studies, 2015 participants; low-certainty evidence).

No data were available for primary nonfunction, acute rejection, quality of life or cardiovascular diseases.

Most studies focused on kidney transplant outcomes, with limited data on other organs. The certainty of the evidence was generally low to moderate, primarily due to imprecision, high risk of reporting bias, or both.

Authors' conclusions

This review provides low-certainty evidence that donor hypothermia may have no impact on post-transplant outcomes. Donor hypothermia may not be associated with reduced organ utilisation. No safety concerns were identified, and donor hypothermia may not increase donor adverse events.

Funding

No specific funding was received.

Registration

Protocol (2023): doi:10.1002/14651858.CD015190

PLAIN LANGUAGE SUMMARY

Does lowering the body temperature of donors following brain death improve the quality of transplanted organs?

Key messages

- Lowering the body temperature of donors in the intensive care unit before retrieving organs may not improve how soon the organ starts to work in the recipient.
- Lowering the body temperature of donors may not reduce the number of organs that could be used for transplant or cause any safety problems (such as heart issues).
- One-year survival of the transplanted organ was only reported by three of the four studies. Quality of life and heart disease for the recipients at 12 months were not reported in any of the studies. Key organ-related measures at 12 months (e.g. heart: heart function; pancreas: the need for insulin) were not reported.

What is solid organ transplantation?

Solid organ transplantation is a surgical procedure where a solid organ (kidney, heart, liver, lung or pancreas) is transferred from one person - the donor - to another person - the recipient. It is the optimal treatment for people living with end-stage organ failure. However, there remains a shortage of optimal organs for donation. Many organs come from people who are brain dead (where a person has permanent and irreversible loss of brain function) in an intensive care unit. The process of brain death is known to cause significant injury to organs. Therefore, efforts to optimise the care of these donors should be maximised to achieve the best outcomes and graft function in the transplant recipients.

Why lower the body temperature of donors, and how is it done?

Lowering the body temperature of donors to below normal core temperature (< 35.5°C) may slow the body's processes, helping to keep the organs healthier until they are retrieved and transplanted. This can take place in the intensive care unit between brain death and organ donation. Lowering the body temperature may be achieved by allowing donors to spontaneously cool to the target temperature, using fans, ice packs, or specific external cooling systems to promote the lowering of the body temperature of the donor, or bladder irrigation, invasive cooling lines or temperature-controlled humidified gases used during ventilation to achieve the desired core temperature. The degree of therapeutic hypothermia can also be classified as mild (34.0°C to 35.5°C), moderate (32.0°C to 33.9°C), moderately deep (30.0°C to 31.9°C) and deep (< 30°C).

What did we want to find out?

We wanted to see whether lowering the body temperature of donors makes organs healthier and improves outcomes for recipients after an organ transplant. We were interested in:

- how soon donated organs started working;
- whether the transplant was successful;
- whether the recipient survived (and for how long).

What did we do?

We searched for studies that assessed whether lowering the body temperature of donors was effective and if it caused unwanted effects. Donors in the studies had to have irreversible loss of all functions of the entire brain or irreversible loss of the capacity for consciousness and an irreversible loss of the ability to breathe. Recipients had to be receiving solid organs such as kidney, heart, liver, lung or pancreas. Studies could be set anywhere in the world.

What did we find?

We found four studies with 2096 donors that looked at the effect of lowering the body temperature of donors prior to transplantation. Two studies looked at kidney transplantation, and two studies looked at multiple solid organs (kidney, liver, lung, heart, pancreas). Studies were undertaken in the USA and France.

Kidney transplantation

Lowering the body temperature of donors compared to keeping them at normal body temperature:

- may not reduce the time it takes for transplanted kidneys to start working (4 studies, 3015 recipients);
- may make no difference to survival of the transplanted kidney one year after transplant (3 studies, 2075 recipients);
- may make no difference to how many kidney recipients survived for a year after transplant (1 study, 526 recipients).

Other organs

Lowering the body temperature of donors compared to keeping them at normal body temperature:

- may not reduce the time for transplanted lungs to start to work (1 study, 99 recipients);
- may not reduce the time for transplanted livers to start to work (1 study, 262 recipients);
- is unlikely to reduce the number of organs available for transplant (4 studies, 4265 recipients).

Lowering the body temperature of donors compared with keeping them at normal body temperature may not cause any problems with the donors (such as heart issues) (3 studies, 2015 donors).

What are the limitations of the evidence?

The small number of studies investigated, and the donor organs transplanted were the major limitations of this review. One-year survival of the transplanted organ was only reported by three studies. Quality of life and heart disease for the recipients at 12 months were not reported in any of the studies. Key organ-related measures at 12 months (e.g. heart: heart function; pancreas: the need for insulin) were not reported. Most studies focused on kidney donors and recipients, with very little information for other solid organs.

How up to date is this evidence?

The review is current to September 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Donor hypothermia compared to donor normothermia for recipients of kidneys from deceased donors following brain death

Donor hypothermia compared to donor normothermia for recipients of kidneys from deceased donors following brain death

Patient or population: recipients of kidneys from deceased donors following brain death
Setting: hospital intensive care units
Intervention: donor hypothermia
Comparison: donor normothermia

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with donor normothermia	Risk with donor hypothermia				
Delayed graft function follow-up: median 7 days	230 per 1000	**200 per 1000** (164 to 249)	**RR 0.87** (0.71 to 1.08)	3015 (4 RCTs)	⊕⊕⊝⊝ Low[a,b]	
Kidney 1-year graft survival follow-up: median 12 months	Moderate		**HR 0.77** (0.51 to 1.14) [Graft loss]	2075 (4 RCTs)	⊕⊕⊝⊝ Low[a,b]	
	100 per 1000	**170 per 1000** (72 to 309)				
Kidney 1-year patient survival follow-up: median 12 months	Moderate		**HR 0.81** (0.42 to 1.57) [Death]	526 (1 RCT)	⊕⊕⊝⊝ Low[c]	
	100 per 1000	**155 per 1000** (27 to 380)				
Kidney utilisation	808 per 1000	**800 per 1000** (767 to 832)	**RR 0.99** (0.95 to 1.03)	4265 (4 RCTs)	⊕⊕⊕⊝ Moderate[a]	
Donor adverse events	9 per 1000	**12 per 1000** (4 to 34)	**RR 1.28** (0.45 to 3.62)	2015 (3 RCTs)	⊕⊕⊝⊝ Low[c]	
Kidney primary graft non-function	0 per 1000	**-- per 1000** (-- to --)	Not estimable	(0 RCTs)	-	
Kidney acute rejection	0 per 1000	**-- per 1000** (-- to --)	Not estimable	(0 RCTs)	-	

*The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **HR:** hazard ratio; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_460070498252499600.

[a] Downgraded for risk of bias: Malinoski 2023 was assessed as having a high risk of bias associated with the outcome.
[b] Downgraded for imprecision: the CIs are wide and include both important benefit and harm.
[c] Downgraded for imprecision twice: the CIs are very wide, and include both benefit and harm. The number of events is very small, making the estimate unreliable.

Summary of findings 2. Summary of findings table - Donor hypothermia compared to donor normothermia for recipients of lung, liver, heart or pancreas from deceased donors following brain death

Donor hypothermia compared to donor normothermia for recipients of lung, liver, heart or pancreas from deceased donors following brain death

Patient or population: recipients of lung, liver, heart or pancreas from deceased donors following brain death
Setting: hospital intensive care units
Intervention: donor hypothermia
Comparison: donor normothermia

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with donor normothermia	Risk with donor hypothermia				
Lung 1-year graft survival follow-up: median 12 months	Moderate		**HR 0.73** (0.21 to 2.51) [Graft loss]	99 (1 RCT)	⊕⊕⊝⊝ Low[a]	
	100 per 1000	**186 per 1000** (3 to 617)				
Liver 1-year graft survival follow-up: median 12 months	Moderate		**HR 0.85** (0.35 to 2.06) [Graft loss]	262 (1 RCT)	⊕⊕⊝⊝ Low[a]	
	100 per 1000	**141 per 1000** (9 to 447)				

Pancreas 1-year graft survival follow-up: median 12 months	Moderate		Not estimable	(0 RCTs)	-
	100 per 1000	**-- per 1000** (-- to --)			
Heart 1-year graft survival follow-up: median 12 months	Moderate		Not estimable	(0 RCTs)	-
	0 per 1000	**-- per 1000** (-- to --)			
Lung utilisation	267 per 1000	**259 per 1000** (227 to 297)	**RR 0.97** (0.85 to 1.11)	2518 (2 RCTs)	⊕⊕⊕⊝ Moderate[b]
Liver utilisation	774 per 1000	**789 per 1000** (743 to 836)	**RR 1.02** (0.96 to 1.08)	1259 (2 RCTs)	⊕⊕⊕⊕ High
Pancreas utilisation	75 per 1000	**81 per 1000** (55 to 120)	**RR 1.09** (0.74 to 1.61)	1259 (2 RCTs)	⊕⊕⊕⊝ Moderate[b]
Heart utilisation	307 per 1000	**316 per 1000** (267 to 371)	**RR 1.03** (0.87 to 1.21)	1259 (2 RCTs)	⊕⊕⊕⊝ Moderate[b]

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **HR:** hazard ratio; **RR:** risk ratio

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_467353674397915395.

[a] Downgraded for imprecision twice: the CIs are very wide, and include both benefit and harm. The number of events is very small, making the estimate unreliable.
[b] Downgraded for imprecision: the CIs are wide and include both important benefit and harm.

BACKGROUND

Description of the condition

Solid organ transplantation is the optimal treatment for patients living with end-stage organ failure. However, there remains a shortage of optimal organs for donation.

The process of brain death is known to cause significant injury to organs [1, 2, 3, 4]. Therefore, efforts to optimise the perioperative care of these donors should be maximised to achieve the best outcomes and graft function in transplant recipients. These improved outcomes may also facilitate increased use of organs from expanded criteria donors, therefore, growing the donor pool. Despite these potential benefits, few interventions have been researched in this population.

Description of the intervention and how it might work

Standard clinical practice is to maintain normothermia (a core body temperature between 36°C and 38°C) following brainstem death up to the point of organ donation. As thermoregulation becomes impaired following brain death (due to loss of thalamic and hypothalamic function, peripheral vasodilation and reduced metabolic rate), spontaneous donor hypothermia is not uncommon, frequently requiring rewarming interventions to maintain normothermia [3, 5, 6].

Therapeutic donor hypothermia is a process in which donors are deliberately allowed to reach a core body temperature of < 35.5°C before organ retrieval [7]. This may be achieved by passive methods (allowing donors to spontaneously cool to their target temperature), active non-invasive methods (such as using fans, ice packs, or specific external cooling systems to promote cooling), or active invasive methods (such as bladder irrigation, invasive cooling lines, or temperature-controlled humidified gases during invasive ventilation) to achieve the desired core temperature. The degree of therapeutic hypothermia can also be classified into mild (34.0°C to 35.5°C), moderate (32.0°C to 33.9°C), moderately deep (30.0°C to 31.9°C) and deep (< 30°C) [8]. Importantly, this differs from accidental hypothermia, which is commonly due to environmental factors leading to a reduced core body temperature in patients before their arrival in hospital and has a different severity grading [9].

Therapeutic hypothermia is classically divided into three stages: induction, maintenance and rewarming [10]. Given the study population of interest in this review, it is unlikely that the rewarming phase will be used during their clinical care. The optimal depth, duration and method of achieving therapeutic hypothermia is debated and may vary according to its clinical indication. It is commonly employed as a neuroprotective strategy in patients following cardiac arrest and in children with hypoxic ischaemic encephalopathy [11]. In addition, there is considerable ongoing research to establish its efficacy in the care of patients with refractory status epilepticus, hepatic encephalopathy, traumatic brain injury and acute ischaemic stroke [11, 12, 13, 14].

Hypothermia has complex and profound physiologic implications on the body. Notably, it reduces metabolism even with moderate hypothermia between 32°C and 34°C, resulting in a 50% to 60% reduction in basal metabolic rate [8, 10]. It also leads to reduced oxygen-derived free radical production and less oxidative stress on tissues. While these mechanisms have primarily been used for neuroprotection, they may confer similar protection on other organs that may later be donated.

This is particularly important given the physiologic changes that occur following brain death, notably with changes in vasomotor tone leading to dramatic swings in blood pressure and occasionally cardiovascular collapse [3]. Before this derangement can be corrected by clinicians, the resulting hypoperfusion may lead to tissue damage, which could impair the function of later donated organs and, therefore, worsen outcomes for recipients. The effects of therapeutic hypothermia are postulated to confer some protection against these physiologic insults.

Similarly, hypothermia has been shown in other clinical settings to protect against ischaemia-reperfusion injury [13, 15, 16, 17]. Ischaemia-reperfusion injury is inevitable in transplantation, and the duration of both cold and warm ischaemic times is a key factor determining graft outcome [18, 19, 20]. As such, it is hoped that therapeutic donor hypothermia could reduce the effects of ischaemia-reperfusion injury and improve recipient outcomes.

Why it is important to do this review

Every effort should be made to increase the number of suitable organs available for transplantation, given the large number of patients awaiting organ transplantation and the paucity of available optimal organs. Despite this, research into specific interventions to improve the quality of organs prior to donation in donors following brain death is scarce [2].

Therapeutic hypothermia is an intervention with a growing evidence base to support its use in clinical practice, particularly for its effects on preventing tissue damage from oxidative stress and ischaemia-reperfusion injury. However, it is not without risk, and the adverse effects of hypothermia range from increased risk of infection to fatal arrhythmia and cardiac arrest [3]. It also places demands on critical care nursing staff, which are appropriate only if the intervention demonstrates benefit [21]. When applied to organ donation, therapeutic hypothermia in donors following brain death may increase the quality of transplanted organs by conferring protection from hypoperfusion and ischaemia-reperfusion injury, ultimately leading to improved graft function and better patient-centred outcomes for recipients.

Even within its current recognised indications, there is ongoing debate about the optimal implementation strategy, target temperature, method of cooling and duration of therapy, with a constantly evolving evidence base to help refine best practices [10, 22, 23].

Therefore, this review is essential to assess the extent and quality of current evidence to inform whether therapeutic hypothermia is safe and efficacious in improving the quality of transplanted organs in donors following brain death.

OBJECTIVES

To assess the benefits and harms of therapeutic donor hypothermia in recipients of organs donated from brain-dead donors.

METHODS

Differences between protocol and review

In our initial protocol, subgroup analysis based on machine perfusion use was not pre-specified. This subgroup analysis has been performed because machine perfusion is likely to influence the quality of the organs and the likelihood of adverse outcomes.

Although we planned to create a summary of findings (SoF) table for each organ type, we created one for kidney transplantation and one combined SoF table for all other organ types.

Criteria for considering studies for this review

Types of studies

All randomised controlled trials (RCTs) and quasi-RCTs (RCTs in which allocation to treatment was obtained by alternation, use of alternate medical records, date of birth or other predictable methods) looking at therapeutic donor hypothermia following brainstem death to improve the quality of transplanted organs.

Types of participants

Inclusion criteria

For donors

All organ donors who have received a formal diagnosis of brain death, regardless of age, ethnicity or sex.

As the criteria for diagnosing death by neurological criteria can vary between countries [24], for the purposes of this review, brain death was considered to be any that satisfies the recommendations for the determination of brain death outlined by the World Brain Death Project [25]. This is a clinical examination that demonstrates coma, brainstem areflexia and apnoea in patients with an established neurological diagnosis that can lead to irreversible loss of brain function and where confounders of clinical examination and mimics of brain death have been excluded.

Practically, this includes both: 1) whole brain death, defined as the irreversible cessation of all functions of the entire brain, including the brainstem, or 2) brainstem death, defined as the irreversible loss of the capacity for consciousness and the irreversible loss of the ability to breathe, and these are the criteria that are used throughout the USA, the UK and most European countries.

For recipients

Receiving one of the following organs regardless of age, ethnicity, or sex: heart, lung, liver, kidney, pancreas.

Exclusion criteria

For donors

Donation following circulatory death, as donor hypothermia would be an unacceptable pre-mortem intervention in this cohort.

For recipients

No specific exclusion criteria.

Types of interventions

Hypothermia

This review included passive, active non-invasive, active invasive, or any combination of these methods for achieving therapeutic donor hypothermia.

Passive

- Allowing donors to spontaneously reach their target temperature.

Active non-invasive

- Fans, ice packs, adjusting room temperature, removing clothing, blankets or both
- Water circulating blankets (e.g. Blanketrol®): these are water-filled blankets placed on the patient's body that circulate temperature-controlled water continuously
- Air-circulating blankets (e.g. 3M™ Bair Hugger™ System): these blankets deliver temperature-controlled air through a blanket placed over the patient.
- Water-circulating gel-coated pads (e.g. Arctic Sun®): these use adhesive gel pads placed directly on the patient's body, with temperature-controlled water circulating through the pads.

Active invasive

- Intravascular heat exchange system
- Bladder irrigation
- Peritoneal irrigation
- Pleural irrigation.

Other considerations

- The depth of hypothermia (mild, moderate, moderately deep or deep)
- How the target temperature was measured (e.g. oesophageal, rectal or tympanic thermometers and whether the temperature was monitored continuously or intermittently)
- The duration of hypothermia prior to organ retrieval
- The time at which the target temperature was achieved
- Whether the target temperature was or was not achieved
- The setting in which these interventions were being delivered prior to organ retrieval. While we expect donors receiving therapeutic hypothermia to be almost exclusively cared for in critical care units, it is possible that they may also receive these interventions in emergency departments, operating theatres or other clinical settings.

Normothermia

The comparator is donor normothermia, which is a standard clinical practice where the core donor body temperature is maintained between 36°C and 38°C following brainstem death up to the point of organ donation.

Outcome measures

The outcomes selected have been based on expert opinion, previous literature in the field and the SONG core outcome sets for transplantation as specified by the Standardised Outcomes in Nephrology Initiative [26].

Critical outcomes

- Post-transplant outcomes
  - Delayed graft function (DGF). For kidney transplants, this is defined as the need for dialysis within seven days post-transplant.
  - Death-censored graft survival (for all organ transplants): time from transplant to graft failure, censoring for patient death (reported as time-to-event data at one-year or as reported)
- Utilisation: number of organs transplanted out of the total available organs
- Donor adverse events, including but not limited to:
  - Allograft thrombosis
  - Cancer
  - Infection
  - In-hospital death
  - Severe arrhythmia and cardiac arrest
  - Liver disorders: measured using alanine aminotransferase (ALT), aspartate aminotransferase (AST), gamma-glutamyl transferase (GGT), alkaline phosphatase (ALP) and bilirubin
  - Thrombocytopenia
  - Requirement for organ support for organs other than those transplanted.

Important outcomes

For all organ recipients

- **Patient survival**: time from transplant to death from any cause (reported as time-to-event data at one year, or as reported by the study)
- **Primary nonfunction**: immediate and irreversible failure of the transplanted organ following transplantation
- **Acute rejection**: immune-mediated injury to transplanted organs within weeks to months following transplantation. This will include biopsy-proven and treated acute rejection.
- **Quality of life (QoL) and life participation**: potential measurements include, but not limited to:
  - World Health Organisation Quality of Life assessment (WHOQoL) [27]
  - Kidney Disease Quality of Life (KDQoL) [28]
  - Other validated instruments, where WHOQoL or KDQoL were not reported.
- **Cardiovascular diseases**, including but not limited to:
  - Coronary artery disease
  - Heart failure
  - Cerebrovascular disease.

Organ-specific outcome measures

- **Kidney**: estimated glomerular filtration rate (eGFR) at one year and at the maximum follow-up period.
- **Liver**: early allograft dysfunction (seven days) and ischaemic type biliary complications at one year and at the maximum follow-up period.
- **Pancreas**: assessed primarily using the Igls consensus criteria: glycated haemoglobin (HbA1c), occurrence of severe hypoglycaemia, > 50% reduction in insulin requirements, and increase in C-peptide levels [29]. We will also include within one year or the maximum follow-up period.
- **Lung**: graft function measured using the pulmonary graft dysfunction (PGD) score at one year or at the maximum follow-up period.
- **Heart**: left ventricular ejection fraction, cardiac allograft vasculopathy, and New York Heart Association (NYHA) score at one year or at the maximum follow-up period.

Search methods for identification of studies

Electronic searches

The Cochrane Kidney and Transplant Information Specialist searched the following databases for RCTs without language, publication year or publication status restrictions.

- Cochrane Kidney and Transplant Specialised Register via MeerKat (a software application built on Microsoft Access used to manage the database). MeerKat is a study-based register and is used to store bibliographic and study details, link multiple reports to a single study, link studies to reviews, and track the progress of reviews (3 September 2025).
- Cochrane Central Register of Controlled Trials (CENTRAL) via the Cochrane Library (Issue 9, 2025)
- Ovid MEDLINE® ALL (from 1946 to 3 September 2025)
- Embase.com records as part of the search of CENTRAL as described in How CENTRAL is created (https://www.cochranelibrary.com/central/central-creation)
- US National Institutes of Health Ongoing Trials Register ClinicalTrials.gov (https://www.clinicaltrials.gov) as part of the search of CENTRAL
- World Health Organization International Clinical Trials Registry Platform (https://trialsearch.who.int) as part of the search of CENTRAL.

The Information Specialist modelled the subject-specific strategies for other databases on the search strategy designed for MEDLINE. MEDLINE searches combine the subject strategy adaptations with the Sensitivity and precision maximising search strategy designed by Cochrane for identifying RCTs (as described in the *Cochrane Handbook for Systematic Reviews of Interventions* Version 5.1.0) and updated by us to account for new MeSH study types and wider use of machine indexing by the National Library of Medicine (NLM) [30]. Additional search strategies have been designed by the Information Specialist for MEDLINE, Embase and CENTRAL.

See Supplementary material 1 for search terms used in strategies for this review.

Date of search: 3 September 2025

Searching other resources

1. We checked the reference lists of included studies and any relevant systematic reviews identified for further references to relevant studies.
2. We checked the included studies for retractions and errata via the Retraction Watch Database and report the search dates in the review (https://retractiondatabase.org).
3. We searched Epistemonikos for related systematic reviews (https://www.epistemonikos.org).
4. We contacted the original authors or funders of the included studies for clarification and further data, but we did not get any response.

Data collection and analysis

Selection of studies

The search strategy described was used to retrieve titles and abstracts of studies potentially relevant to the review. The titles and abstracts were independently screened by two review authors (TH, ST), who discarded studies that were not applicable; however, studies and reviews that were thought to include relevant data or information on studies were initially retained. Two review authors independently assessed retrieved abstracts (TH, ST) and, if necessary, the full text of these studies to determine which studies satisfied the inclusion criteria.

Data extraction and management

Data extraction was conducted independently by two review authors (TH, ST) using standardised data extraction forms. Disagreements were planned to be resolved in consultation with a third review author (CW). Studies reported in non-English language journals were planned to be translated before assessment, but no relevant studies were identified. Where more than one publication of a study existed, reports were grouped together, and the publication with the most complete data was planned to be used in the analyses. Where relevant outcomes are only published in earlier versions, these data were planned to be used. Any discrepancies between published versions we planned to highlight.

Risk of bias assessment in included studies

The following items were independently assessed by two review authors (TH, ST) using the risk of bias assessment tool 1 [31] (see Supplementary material 6).

- Was there adequate sequence generation (selection bias)?
- Was allocation adequately concealed (selection bias)?
- Was knowledge of the allocated interventions adequately prevented during the study?
  - Participants and personnel (performance bias)
  - Outcome assessors (detection bias)
- Were incomplete outcome data adequately addressed (attrition bias)?
- Are reports of the study free of suggestion of selective outcome reporting (reporting bias)?
- Was the study apparently free of other problems that could put it at risk of bias?

Measures of treatment effect

For dichotomous outcomes (e.g. DGF, primary graft nonfunction), results were expressed as risk ratios (RR) with 95% confidence intervals (CI). Where continuous measurement scales were used to assess treatment effects (e.g. eGFR, length of stay), the mean difference (MD) was used.

We analysed graft survival and patient survival as time-to-event data and performed meta-analyses of log hazard ratios (HR) using the general inverse-variance method, as described in Chapter 10 of the *Cochrane Handbook for Systematic Reviews of Interventions* [32]. Where HRs were reported (with CIs, P values, or both), these were used directly. Where a study performed time-to-event analyses but did not report an HR, we planned to impute HRs and 95% CIs using the techniques described by Tierney and colleagues [33].

If available data in included studies were insufficient to allow time-to-event analysis, we planned to analyse survival as a dichotomous variable at set time points (one, three, and five years). In this instance, we planned to use one-year graft survival as a primary outcome.

Where standard deviations (SD) were not given, we planned to directly obtain them from available statistics (e.g. standard errors) using the RevMan Calculator [34]. When only the median and interquartile range (IQR) were given, we planned to use methods described in [35] to estimate the mean and SD. If this was not possible, SDs were planned to be imputed from other studies [36].

Donor adverse events were assessed as risk differences with 95% CIs for each compared to no treatment (donor normothermia). These were also planned to be tabulated and assessed with descriptive techniques, but were not performed due to insufficient data.

Unit of analysis issues

We did not anticipate any unit of analysis issues, as we did not expect any cluster-RCTs or cross-over RCTs, or studies with more than two intervention arms relevant to this review, to be included in this review. If such studies are present in future versions of this review, we will address these as described in [37].

Dealing with missing data

Any further information required from the original author was requested by written correspondence (e.g. emailing the corresponding author). Any relevant information obtained in this manner was to be included in the review. Evaluation of important numerical data, such as screened, randomised patients, as well as intention-to-treat, as-treated and per-protocol population, were carefully performed. Attrition rates (i.e. dropouts, losses to follow-up, withdrawals) were investigated. Issues of missing data and imputation methods (e.g. last-observation-carried-forward) were critically appraised [32].

Reporting bias assessment

Funnel plots were planned to assess small-study effects in the event of having more than 10 studies. However, they were not used because only four studies were included in the meta-analyses [38].

Synthesis methods

Data were pooled using the random-effects model as we expected there would be clinical diversity. The DerSimonian and Laird estimator was used to estimate between-trial variance, and the Wald-type method was used to calculate a CI for the meta-analysis effect estimate [32]. The fixed-effects model was also used to assess the robustness of the chosen model and its susceptibility to outliers.

Investigation of heterogeneity and subgroup analysis

We assessed for clinical heterogeneity by documenting variability in participants, interventions and outcomes of the included trials. We analysed methodological heterogeneity by evaluating how studies conducted randomisation, blinding, loss to follow-up and intervention type. Statistical heterogeneity was quantified using the I^2 statistic, which describes the percentage of total variation across studies that is due to heterogeneity rather than sampling error [39]. A guide to the interpretation of I^2 values is as follows.

- 0% to 40%: might not be important
- 30% to 60%: may represent moderate heterogeneity
- 50% to 90%: may represent substantial heterogeneity
- 75% to 100%: considerable heterogeneity.

The importance of the observed value of I^2 depends on the magnitude and direction of treatment effects and the strength of evidence for heterogeneity (e.g. P value from the Chi^2 test or a CI for I^2) [32].

Additionally, we performed the following subgroup analyses.

- Standard criteria donors (SCD) versus extended criteria donors (ECD), as this was likely to influence the quality of organs and the likelihood of adverse outcomes. ECD are defined as those > 60 years or aged 50 to 59 years and having at least two additional risk factors (history of hypertension, creatinine >132 µmol/L, or cerebrovascular cause of death). SCD are defined as donors aged < 60 years who do not meet the ECD criteria.
- For similar reasons as above (variability in organ quality/ injury), we also performed subgroup analyses based on machine perfusion.
- Studies with mean/median time at target temperature above versus below 12 hours. No study aimed to achieve any minimum duration of cooling (e.g. 12 hours) prior to organ retrieval.

The following subgroup analyses were planned, but were not performed due to insufficient data.

- Mild versus moderate, versus moderately deep versus deep hypothermia may have different physiological effects, which can influence treatment outcomes.
- Passive versus active non-invasive versus active invasive methods of achieving hypothermia, as risks and benefits may differ in these groups.
- Rapidly achieved target temperature (e.g. within four hours of initiation of therapy) compared to cases where a longer time to achieve target temperature was observed.
- An analysis of only organs not subjected to prolonged cold and warm ischaemic times, as this is known to affect graft outcomes.
- Organs that were immunologically well-matched compared to those that were not immunologically well-matched, in case any benefits or adverse effects were not observed in either group.

Equity-related assessment

We assessed the country of origin of the included studies and classified them as low-, middle- or high-income countries (The World Bank).

Sensitivity analysis

We performed sensitivity analyses to explore the influence of the following factors on effect size.

- Repeating the analysis, taking account of the risk of bias, as specified
- Repeating the analysis based on the country.

The following sensitivity analyses were planned but not performed due to insufficient data.

- Repeating the analysis, excluding unpublished studies
- Repeating the analysis, excluding any very long or large studies to establish how much they dominate the results
- Repeating the analysis, excluding studies using the following filters: diagnostic criteria, language of publication, source of funding (industry versus other).

Certainty of the evidence assessment

We presented the main results of the review in SoF tables. These tables present key information concerning the certainty of the evidence, the magnitude of the effects of the interventions examined, and the sum of the available data for the main outcomes [40]. The SoF tables also include an overall grading of the evidence related to each of the main outcomes using the GRADE approach [41, 42]. The GRADE approach defines the certainty of a body of evidence as the extent to which one can be confident that an estimate of effect or association is close to the true quantity of specific interest. The certainty of a body of evidence involved consideration of the within-study risk of bias (methodological quality), directness of evidence, heterogeneity, precision of effect estimates and risk of publication bias [43].

The following outcomes were included in the SoF tables.

- **Kidney**
  - DGF
  - Graft survival
  - Patient survival
  - Utilisation
  - Donor adverse events
  - Primary graft non-function
  - Acute rejection
- **Other organs (lung, liver, heart, pancreas)**
  - Graft survival
  - Patient survival
  - Organ utilisation

Consumer involvement

Consumers were not involved in this review.

RESULTS

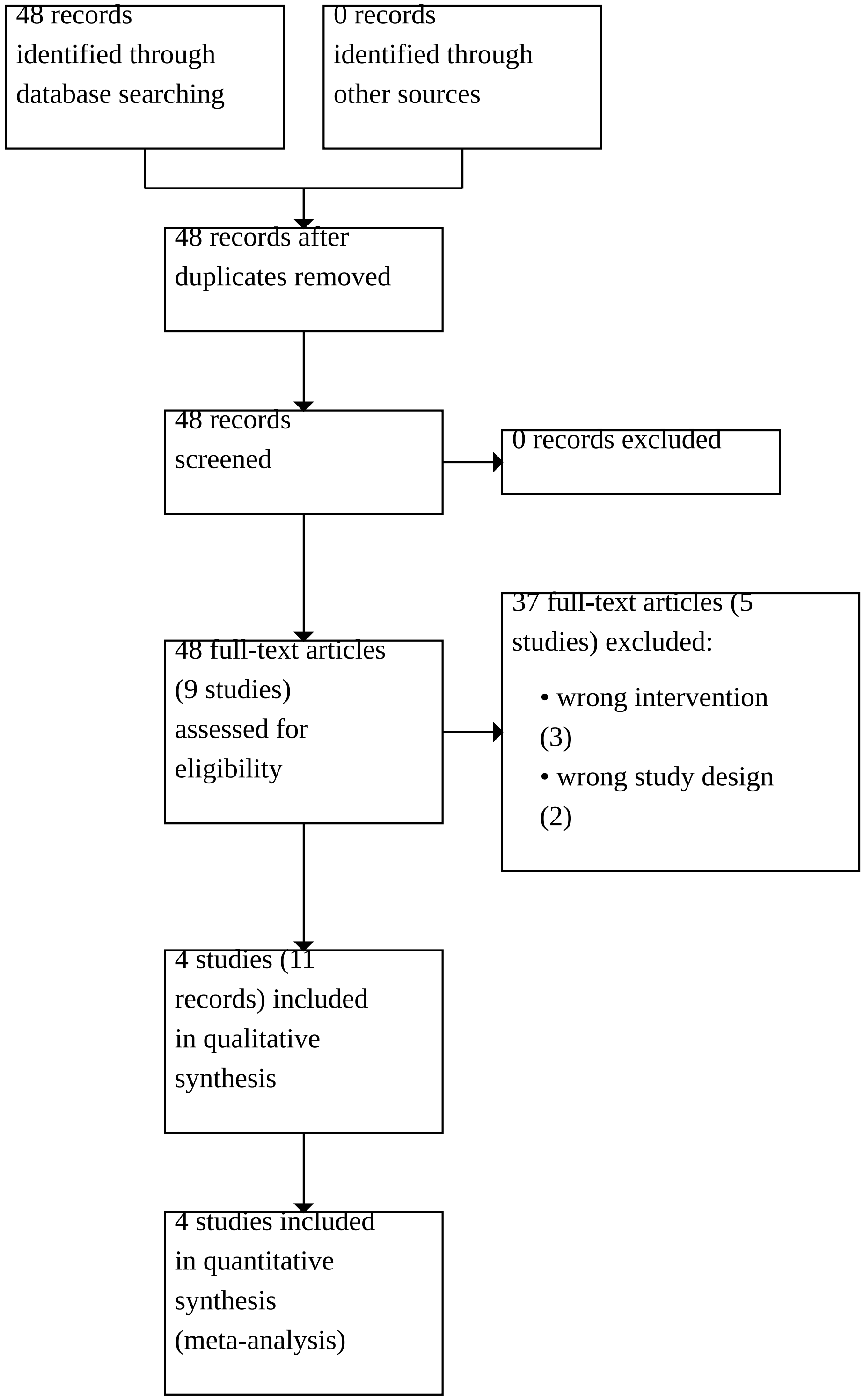
Description of studies

The following section contains broad descriptions of the studies considered in this review. For further details on each individual study, please see the characteristics of included and excluded studies tables (Supplementary material 2; Supplementary material 3).

Results of the search

A flow chart of the study selection can be found in Figure 1.

Figure 1. Flow chart showing study selection



The search identified 48 records. No duplicates were identified. After screening the remaining titles and abstracts and full-text review, four studies (11 records) were included, and five studies (37 records) were excluded. No ongoing studies were identified.

Included studies

Table 1 is a summary of included studies and syntheses, and full details for each study can be found in the 'Characteristics of included studies' (Supplementary material 2). Additional information was requested from the authors of Malinoski 2023 [44, 45], but no information was received.

Patients

This review included organ grafts from 2096 donors across four different multicentre RCTs.

- HYPOREME 2024 [46, 47] was conducted in France. Only ECD were included in the trial. Machine perfusion for kidneys was routinely used.
- Patel 2024 [48] was conducted in the USA. Both ECD and SCD were included in the trial. This study included non-pump-eligible kidney donors.
- Malinoski 2023 was conducted in the USA. Both ECD and SCD were included in the trial. Only donors eligible for their kidney grafts to be preserved using machine perfusion were included in the study, and the eligibility criteria differed between centres.
- Niemann 2015 [49, 50, 51, 52, 53, 54] was conducted in the USA. ECD and SCD were included in the study. Machine perfusion was not used for kidney grafts.

Interventions

Across all included studies, 1125 donors were randomised to receive donor hypothermia, defined as a target core donor temperature of 34ºC to 35ºC. Core body temperature was measured using a bladder probe, intravascular catheter, or an oesophageal probe with or without a rectal probe.

Donor hypothermia was achieved using various methods, including 3M™ Bair Hugger™ System, Arctic Sun®, Blanketrol®, ice packs and fan, cooling blanket, and passive air. All studies aimed to achieve the target temperature within four hours of enrolment, although the duration of hypothermia varied substantially (HYPOREME 2024 (mean ± SD): 2.8 ± 3.8 hours; Patel 2024: not available; Malinoski 2023 (mean ± SD): 30.3 ± 17.0 hours; Niemann 2015 (median, IQR): 16.9 hours, 10.2 to 29.4).

Comparators

A total of 971 donors were randomised to donor normothermia, with a target temperature of 36.5ºC to 37.5ºC. Temperature monitoring methods were similar to those used in the hypothermia groups.

Outcomes

All four studies analysed kidney transplant outcomes, while two studies (Malinoski 2023; Niemann 2015) investigated other solid organ transplant outcomes (heart, lung, liver, pancreas).

Excluded studies

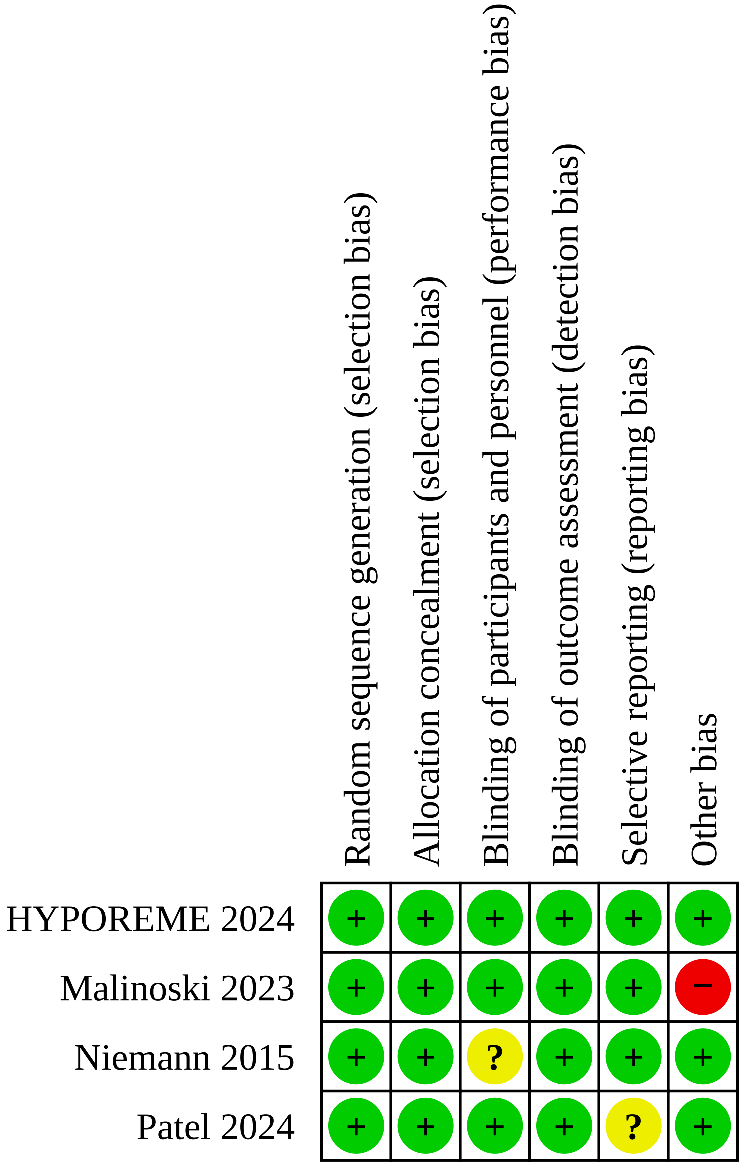
We excluded five studies for the following reasons: two studies did not include donor hypothermia as a randomised intervention; one study assessed the role of additional normothermic machine perfusion versus hypothermic machine perfusion, with no use of therapeutic donor hypothermia; one study compared machine perfusion to static cold storage; and one study assessed the impact of ex-situ machine perfusion on outcomes and not donor hypothermia.

For more details, see 'Characteristics of excluded studies' (Supplementary material 3).

Risk of bias in included studies

The following section provides an overview of some of the common biases in the included studies. For further details on each individual study, please see 'Characteristics of included studies' (Supplementary material 2). Figure 2 summarises the risk of bias items for each study.

Figure 2.



HYPOREME 2024 was assessed as having a low risk of bias across all domains. Random sequence generation and allocation concealment were computer-generated and appear adequate. Outcome assessment was conducted by nephrologists at the transplant centres, who were masked to group allocation as were the kidney recipients, minimising detection bias. All outcomes were pre-specified in a published protocol. A higher-than-expected dropout rate following the interim analysis led to an increase in the enrolment target, but this was managed prospectively and deemed to be appropriate.

Patel 2024 was assessed as having an unclear risk of reporting bias, as the study reported only on kidney DGF, with no data on kidney graft survival and other prespecified secondary outcomes at one year. The study concluded on 21 May 2020, and the last recipient follow-up data for one-year function was collected on 14 June 2021. Despite the collection of one-year follow-up data, secondary outcomes at one year, which were pre-specified in the protocol, were not reported. Random sequence generation and allocation concealment were computer-generated and appear adequate.

Malinoski 2023 was assessed as having a high risk of reporting bias. About 27% (269/989) of kidneys allocated to machine perfusion were not perfused. The study does not report how these deviations were distributed across the hypothermia and normothermia study groups, nor whether the kidneys differed in baseline risk. Additionally, 429 kidneys were excluded post-randomisation and pre-transplantation, and the full breakdown of which groups these 429 kidneys had been assigned to was not provided. Therefore, it is not possible to know whether the non-use rates were balanced between the hypothermia and normothermia groups. Random sequence generation and allocation concealment were computer-generated and appear adequate.

Niemann 2015 was assessed as having a low risk of bias in most areas. Random sequence generation and allocation concealment were appropriately handled with computer-generated block randomisation, and recipient characteristics were well balanced between groups. Cold ischaemia time was significantly shorter in the donor hypothermia group. Although the authors adjusted for the effects of cold ischaemia time, this introduces an unclear risk of performance bias, as clinicians may have consciously or unconsciously provided different care to donors (e.g. quicker transplantation or other differences in perioperative care). Outcome assessment was considered low risk for the main endpoints (DGF, graft survival, recipient survival), and outcomes were prespecified in a published protocol. The trial was stopped early for efficacy after a pre-planned interim analysis, which was deemed to be appropriate.

Synthesis of results

A complete record of all comparisons and analyses is available for this review in Supplementary material 4 and a full data package is available in Supplementary material 5.

For kidney outcomes, see Summary of findings 1 for other solid organs, see Summary of findings 2.

Critical outcomes

Post-transplant outcomes

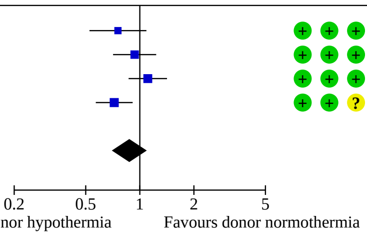
Kidney

All included studies reported kidney DGF as the primary outcome, which was defined as the requirement for dialysis during the first seven days after transplantation. The use of donor hypothermia may not reduce the incidence of DGF compared to donor normothermia (RR 0.87, 95% CI 0.71 to 1.08; I^2 = 57%; 4 studies, 3015 participants; low-certainty evidence; Figure 3).

Figure 3.

	Donor hypothermia		Donor normothermia			Risk Ratio	Risk of Bias					
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Random, 95% CI	A	B	C	D	E	F
HYPOREME 2024	40	251	58	275	19.0%	0.76 [0.52 , 1.09]	+	+	+	+	+	+
Patel 2024	79	460	87	474	25.1%	0.94 [0.71 , 1.23]	+	+	+	+	?	+
Malinoski 2023	103	479	99	510	27.6%	1.11 [0.87 , 1.42]	+	+	+	+	+	−
Niemann 2015	79	280	112	286	28.4%	0.72 [0.57 , 0.91]	+	+	?	+	+	+
Total (Wald[a])		**1470**		**1545**	**100.0%**	**0.87 [0.71 , 1.08]**						
Total events:	301		356									
Test for overall effect: Z = 1.26 (P = 0.21)												

Risk Ratio
M-H, Random, 95% CI



Heterogeneity: Tau² (DL[b]) = 0.03; Chi² = 6.99, df = 3 (P = 0.07); I² = 57%

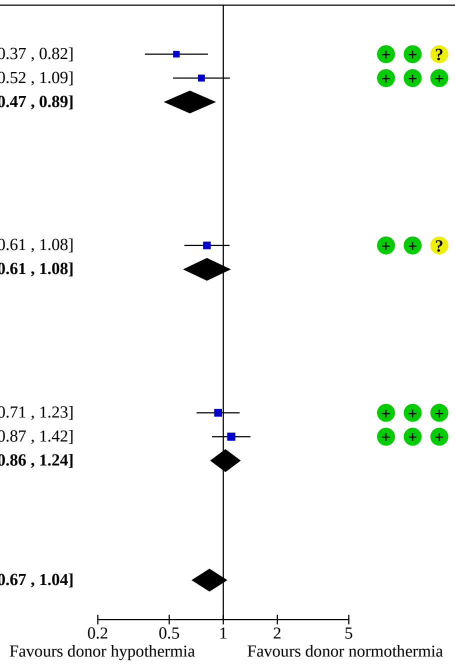
Footnotes
[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Selective reporting (reporting bias)
(F) Other bias

Subgroup analysis was performed based on donor criteria. HYPOREME 2024 only included ECD. Niemann 2015 included both ECD and SCD, and reported outcomes separately for these groups. For Patel 2024 and Malinoski 2023, outcomes were not stratified based on donor criteria and this information was not available. Donor hypothermia may reduce the incidence of DGF among kidney recipients from ECD donors (RR 0.65, 95% CI 0.47 to 0.89; I^2 = 27%; 2 studies, 666 participants; Figure 4; test for subgroup differences Chi^2 = 6.49, df = 2 (P = 0.04), I^2 =69.2%). But no benefit was identified in kidney recipients from SCD donors (RR 0.81, 95% CI 0.61 to 1.08; 1 study, 426 participants; Figure 4). Large subgroup differences were identified between recipients of kidneys from ECD and SCD donors (Analysis 1.2 in Supplementary material 4). Patel 2024 and Malinoski 2023 were added to the mixed donor subgroup, where the majority of recipients received kidneys from SCD donors. This subgroup did not show any benefit with the use of donor hypothermia over normothermia (RR 1.03, 95% CI 0.86 to 1.24; I^2 = 0%; 2 studies, 1923 participants; Figure 4). Subgroup analyses were also performed based on the use of machine perfusion (Analysis 1.3 in Supplementary material 4) and duration of cooling (Analysis 1.4 in Supplementary material 4).

Figure 4.

Study or Subgroup	Donor hypothermia Events	Donor hypothermia Total	Donor normothermia Events	Donor normothermia Total	Weight	Risk Ratio M-H, Random, 95% CI	Risk of Bias A	B	C	D	E	F
1.2.1 Extended criteria donors												
Niemann 2015	22	71	39	69	15.6%	0.55 [0.37 , 0.82]	+	+	?	+	+	+
HYPOREME 2024	40	251	58	275	17.4%	0.76 [0.52 , 1.09]	+	+	+	+	+	+
Subtotal (Wald[a])		**322**		**344**	**33.0%**	**0.65 [0.47 , 0.89]**						
Total events:	62		97									
Test for overall effect: Z = 2.65 (P = 0.008)												
Heterogeneity: Tau² (DL[b]) = 0.01; Chi² = 1.36, df = 1 (P = 0.24); I² = 27%												
1.2.2 Standard criteria donors												
Niemann 2015	57	209	73	217	21.2%	0.81 [0.61 , 1.08]	+	+	?	+	+	+
Subtotal		**209**		**217**	**21.2%**	**0.81 [0.61 , 1.08]**						
Total events:	57		73									
Test for overall effect: Z = 1.42 (P = 0.16)												
Heterogeneity: Not applicable												
1.2.3 Mixed donors												
Patel 2024[c]	79	460	87	474	22.0%	0.94 [0.71 , 1.23]	+	+	+	+	?	+
Malinoski 2023[d]	103	479	99	510	23.8%	1.11 [0.87 , 1.42]	+	+	+	+	+	−
Subtotal (Wald[a])		**939**		**984**	**45.8%**	**1.03 [0.86 , 1.24]**						
Total events:	182		186									
Test for overall effect: Z = 0.29 (P = 0.77)												
Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 0.80, df = 1 (P = 0.37); I² = 0%												
Total (Wald[a])		**1470**		**1545**	**100.0%**	**0.84 [0.67 , 1.04]**						
Total events:	301		356									
Test for overall effect: Z = 1.61 (P = 0.11)												
Test for subgroup differences: Chi² = 6.49, df = 2 (P = 0.04), I² = 69.2%												
Heterogeneity: Tau² (DL[b]) = 0.04; Chi² = 9.82, df = 4 (P = 0.04); I² = 59%												



Footnotes
[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.
[c]98.1% (472) of transplanted kidneys were from SCD donors; 1.9% (9) kidneys were from ECD donors
[d]79.3% (785) of transplanted kidneys were from SCD donors; 20.7% (205) kidneys were from ECD donors

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Selective reporting (reporting bias)
(F) Other bias

Sensitivity analysis was performed excluding Malinoski 2023, which was classified as having a high risk of bias. Here, donor hypothermia reduced the incidence of DGF (Analysis 1.1 in Supplementary material 4). This also led to a decrease in overall heterogeneity (I^2 = 4%, P = 0.35).

All included studies were conducted in high-income countries (France, USA). Sensitivity analysis was carried out by including only studies from the USA (Patel 2024; Malinoski 2023; Niemann 2015), and this did not meaningfully change the main results (Analysis 1.1 in Supplementary material 4).

To ensure the robustness of the model, the main analysis was repeated using a fixed-effect model; donor hypothermia continued to demonstrate no benefit when compared to donor normothermia (Analysis 1.1 in Supplementary material 4).

Only three studies (HYPOREME 2024; Malinoski 2023; Niemann 2015) reported one-year graft survival data as time-to-event analyses. Donor hypothermia may not show any improvement in kidney one-year graft survival (HR 0.77, 95% CI 0.51 to 1.14; I^2 = 0%; 3 studies, 2075 participants; low-certainty evidence; Analysis 1.5).

Other organs

Only Niemann 2015 reported one-year graft survival data as time-to-event analyses for lung and liver transplants. Donor hypothermia may not affect one-year graft survival for lung transplants (HR 0.73, 95% CI 0.21 to 2.51; 1 study; low-certainty evidence; Analysis 2.1) and liver transplants (HR 0.85, 95% CI 0.35 to 2.06; 1 study; low-certainty evidence; Analysis 3.1). Data for heart and pancreas transplants were not reported in the studies due to the small number of events.

Malinoski 2023 reported one-year graft survival data as survival probabilities for liver, heart, lung and pancreas transplants, rather than time-to-event analyses. Overall, one-year graft survival was similar between donor hypothermia and donor normothermia across all organs; survival probabilities: liver (hypothermia 0.97 (0.95 to 0.99), normothermia 0.98 (0.96 to 1.00)); heart (hypothermia 0.98 (0.97 to 1.00), normothermia 0.98 (0.96 to 1.00); lung (hypothermia 0.98 (0.96 to 1.00), normothermia 0.95

(0.91 to 0.99)); and pancreas (hypothermia 0.95 (0.88 to 1.00), normothermia 1.00 (1.00 to 1.00).

Utilisation

Kidney

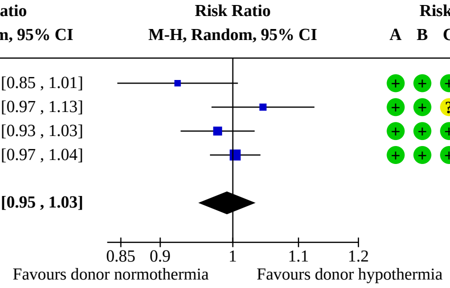
All four studies reported kidney utilisation rates, defined as the number of transplanted kidneys out of the total available kidneys.

Donor hypothermia probably does not affect utilisation rates (RR 0.99, 95% CI 0.95 to 1.03; I^2 = 44%; 4 studies, 4265 participants; moderate-certainty evidence; Figure 5).

Figure 5.

Study or Subgroup	Donor hypothermia Events	Donor hypothermia Total	Donor normothermia Events	Donor normothermia Total	Weight	Risk Ratio M-H, Random, 95% CI	Risk of Bias A	B	C	D	E	F
HYPOREME 2024[a]	258	366	278	364	14.9%	0.92 [0.85 , 1.01]	+	+	+	+	+	+
Niemann 2015[a]	290	360	293	380	18.7%	1.04 [0.97 , 1.13]	+	+	?	+	+	+
Malinoski 2023[a]	837	1114	510	664	27.8%	0.98 [0.93 , 1.03]	+	+	+	+	+	−
Patel 2024	460	500	474	517	38.6%	1.00 [0.97 , 1.04]	+	+	+	+	?	+
Total (Wald[b])		**2340**		**1925**	**100.0%**	**0.99 [0.95 , 1.03]**						
Total events:	1845		1555									

Test for overall effect: Z = 0.43 (P = 0.67)



Heterogeneity: Tau² (DL[c]) = 0.00; Chi² = 5.34, df = 3 (P = 0.15); I² = 44%

Footnotes
[a]Denominator represents the total number of available donor kidneys
[b]CI calculated by Wald-type method.
[c]Tau² calculated by DerSimonian and Laird method.

Risk of bias legend
(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Selective reporting (reporting bias)
(F) Other bias

Subgroup analysis was performed to compare the impact of donor hypothermia based on the use of machine perfusion (Analysis 4.2 in Supplementary material 4). Subgroup analysis based on donor criteria was not possible due to insufficient data.

Other organs

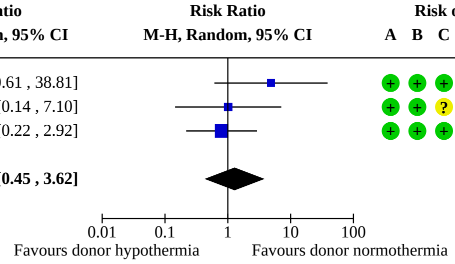
Two studies (Malinoski 2023; Niemann 2015) reported the utilisation data for other organs. Donor hypothermia did not impact utilisation of liver transplants (RR 1.02, 95% CI 0.96 to 1.08; I^2 = 0%; 2 studies, 1259 participants; high-certainty evidence; Analysis 4.4) and probably did not impact utilisation in any of the following organ transplants: heart (RR 1.03, 95% CI 0.87 to 1.21; I^2 = 0%; 2 studies, 1259 participants; moderate-certainty evidence; Analysis 4.3); lung (RR 0.97, 95% CI 0.85 to 1.11; I^2 = 0%; 2 studies, 2518 participants; moderate-certainty evidence; Analysis 4.5); pancreas (RR 1.09, 95% CI 0.74 to 1.61; I^2 = 0%; 2 studies, 1259 participants; moderate-certainty evidence; Analysis 4.6).

Donor adverse events

Three studies (HYPOREME 2024; Malinoski 2023; Niemann 2015) reported the occurrence of donor adverse events. Studies defined donor adverse events based on the following: cardiac arrhythmia/dysrhythmia, cardiac arrest, hypotension, or systemic hypertension. Specific definitions for each study are provided in the footnotes of Analysis 5.1. Donor hypothermia may not affect the occurrence of donor adverse events (RR 1.28, 95% CI 0.45 to 3.62; I^2 = 12%; 3 studies, 2015 participants; low-certainty evidence; Figure 6).

Figure 6.

Study or Subgroup	Donor hypothermia Events	Donor hypothermia Total	Donor normothermia Events	Donor normothermia Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F
Malinoski 2023[a]	8	838	1	510	22.9%	4.87 [0.61 , 38.81]	+	+	+	+	+	−
Niemann 2015[b]	2	150	2	152	25.7%	1.01 [0.14 , 7.10]	+	+	?	+	+	+
HYPOREME 2024[c]	4	183	5	182	51.4%	0.80 [0.22 , 2.92]	+	+	+	+	+	+
Total (Wald[d])		**1171**		**844**	**100.0%**	**1.28 [0.45 , 3.62]**						
Total events:	14		8									
Test for overall effect: Z = 0.47 (P = 0.64)												



Heterogeneity: Tau² (DL[e]) = 0.11; Chi² = 2.27, df = 2 (P = 0.32); I² = 12%

Footnotes

[a]Definition based on cardiac arrhythmia, cardiac arrest, or hypotension
[b]Definition based on dysrhythmia, systemic hypertension, or cardiac arrest
[c]Definition based on cardiac arrhythmia, or cardiac arrest
[d]CI calculated by Wald-type method.
[e]Tau² calculated by DerSimonian and Laird method.

Risk of bias legend

(A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Selective reporting (reporting bias)
(F) Other bias

Important outcomes

Patient survival

Kidney

Only HYPOREME 2024 reported one-year patient survival outcome as time-to-event data. Donor hypothermia may show no difference in one-year patient survival (HR 0.81, 95% CI 0.42 to 1.57; 1 study, 526 participants; low-certainty evidence; Analysis 1.6). Patel 2024 did not report one-year patient survival outcomes. Malinoski 2023 reported patient survival data but not as time-to-event analysis, nor did it report any P values. Niemann 2015 did not have patient survival as one of its outcomes.

Other outcomes

The following were pre-specified as important outcomes in the protocol. However, none of the included studies reported on these measures.

- Primary nonfunction
- Acute rejection
- QoL and life participation
- Cardiovascular disease
- **Kidney**: eGFR at one year and maximal follow-up
- **Liver**: early allograft dysfunction (seven days) and ischaemic type biliary complications at one year and maximal follow-up
- **Pancreas**: this was planned to be assessed primarily using the Igls consensus criteria: HbA1c, occurrence of severe hypoglycaemia, > 50% reduction in insulin requirements, and increase in C-peptide levels [29]. We had also planned to include within one year or the maximum follow-up time.
- **Lung**: graft function measured using the PGD score at one year or the maximum follow-up time.
- **Heart**: left ventricular ejection fraction, cardiac allograft vasculopathy, and NYHA score at one year or the maximum follow-up time.

Equity assessment

All included studies were conducted in high-income countries. No studies were identified from low- or middle-income countries.

Reporting biases

Funnel plots were planned to assess the potential existence of small study bias, but not utilised due to the small number of included studies.

DISCUSSION

Summary of main results

A total of 2096 donors were included across four different studies. Most of the included evidence focused on kidney transplant outcomes, with limited data available on other solid organ transplants. The certainty of the evidence ranged from low to moderate. Two studies (Malinoski 2023; Patel 2024) were at high risk of reporting bias, and many outcomes were limited by imprecision. Table 1 is a summary of included studies and syntheses.

Kidney

All included studies reported the incidence of kidney delayed graft function (DGF); the use of donor hypothermia may not reduce the rate of DGF compared with donor normothermia.

Subgroup analysis revealed that donor hypothermia may reduce DGF incidence in kidney recipients of expanded criteria donors (ECD), while no benefit may be observed in recipients of standard criteria donors (SCD). Although Patel 2024 and Malinoski 2023 were not included in the SCD subgroup, the majority of recipients received kidneys from SCD donors, and this subgroup did not show any benefit with donor hypothermia either. Grafts from ECD donors are typically at a higher risk of DGF, making them more vulnerable to ischaemia-reperfusion injury. Donor hypothermia may mitigate this risk by reducing the metabolic demand, particularly in marginal kidneys such as those from ECD donors as seen in the subgroup analysis (Analysis 1.2).

Subgroup analyses based on the use of machine perfusion (Analysis 1.3) or the duration at target temperature (< 12 hours versus ≥ 12 hours) (Analysis 1.4) did not demonstrate any meaningful changes to the main results. Sensitivity analysis after excluding Malinoski 2023, which was classified as having a high risk of bias, revealed a significant benefit of donor hypothermia in reducing DGF and reduced the heterogeneity among studies (I^2 = 4%, P = 0.35). Sensitivity analysis only including studies conducted in the USA, did not meaningfully alter the results.

For one-year graft survival, donor hypothermia may not demonstrate any impact on outcome. Similarly, donor hypothermia may also not be associated with one-year patient survival.

Donor hypothermia probably has no influence on kidney utilisation rates. Subgroup analysis based on routine use of machine perfusion did not meaningfully change the results (Analysis 4.2).

Other organs

The impact of donor hypothermia on other solid organs was less extensively studied. Only Niemann 2015 reported the impact of donor hypothermia for lung and liver transplants on one-year graft survival as time-to-event analysis. Donor hypothermia may not impact one-year graft survival in lung and liver transplant outcomes.

Malinoski 2023 and Niemann 2015 reported utilisation rates for the following organ grafts: heart, liver, lung, pancreas. Donor hypothermia was probably not associated with utilisation compared with donor normothermia (Analysis 4.3; Analysis 4.4; Analysis 4.5; Analysis 4.6).

Donor adverse events

All included studies reported donor adverse events, defined as cardiac arrhythmia, cardiac arrest, hypotension or systemic hypertension. Donor hypothermia was not associated with donor adverse events, suggesting that the intervention is safe.

Limitations of the evidence included in the review

The primary outcome of all included studies was the incidence of kidney DGF. This outcome was measured in the first week post-transplant, which means that participants were less likely to be lost due to follow-up, and there was virtually no missing data for this outcome. One-year graft survival was reported by only three studies (HYPOREME 2024; Malinoski 2023; Niemann 2015) as time-to-event data. However, only HYPOREME 2024 reported one-year patient survival as time-to-event data. Malinoski 2023 only reported the raw data, without sufficient information to calculate the HR. The authors of Malinoski 2023 were contacted for more information.

Data on the outcomes of non-kidney organ transplants were severely lacking. Only Niemann 2015 provided usable information for one-year graft survival as time-to-event data. Malinoski 2023 only provided survival probabilities, and the authors of Malinoski 2023 were contacted for more relevant information.

High levels of heterogeneity were observed in the analysis of kidney DGF incidence, which we were able to reduce through sensitivity analysis by excluding Patel 2024 and Malinoski 2023 from the main analysis (Analysis 1.1). Subgroup analysis based on donor criteria revealed there may be significant subgroup differences between ECD versus SCD (subgroup differences P = 0.04). Patel 2024 and Malinoski 2023 had the lowest proportion of ECD kidney transplants compared with other studies, which may have contributed to observed heterogeneity.

Limitations of the review processes

We attempted to limit biases at every stage in our review. The search for studies was conducted systematically using the Cochrane Kidney and Transplant Specialised Register. Two independent authors screened the identified studies prior to inclusion in the review. There is always a possibility that we failed to identify some relevant studies. We experienced a limitation in the review process when attempting to obtain additional information from trial investigators. No author responded to our request for more information.

We added a subgroup analysis based on the use of machine perfusion; although our protocol specified subgroup analyses based on the quality of donor organs, it did not specifically mention machine perfusion. This can introduce bias due to multiple comparisons, but we felt this analysis was essential given the mounting evidence of the importance of machine perfusion in kidney transplantation. For the purpose of post-kidney transplant outcomes, we have considered each kidney to be independent and have not accounted for clustering of kidneys within donors; whilst this is the approach of the individual included trials, this may lead to CIs that are biased towards being too narrow. Individual participant data were not available to assess these potential clustering effects.

Agreements and disagreements with other studies or reviews

Miranda and colleagues (2025) was a systematic review which also compared the impact of donor hypothermia versus donor normothermia in kidney transplants [55]. The results of this review are consistent with their findings. They did not stratify results for hypothermia based on short versus long duration at target temperature. Furthermore, the review did not assess graft and patient survival data as time-to-event analysis. No other systematic reviews on this topic were identified.

AUTHORS' CONCLUSIONS

Implications for practice

This Cochrane review provides low-certainty evidence that donor hypothermia may not improve post-transplant outcomes in the recipient. Donor hypothermia is probably not associated with reduced organ utilisation. Donor hypothermia may not be associated with increased donor adverse events.

Equity-related implications for practice

All studies were conducted in high-income countries, which may limit the applicability of these findings to low- or middle-income countries, where resources and access to preservation technologies may differ.

Implications for research

Future work should also be performed with involvement from patients and the public to define optimal outcomes for transplantation studies. Among the included studies, delayed graft function was used as the standard primary outcome for assessing kidney transplant outcomes, with limited reporting of graft survival, patient survival, or other markers of quality of life. Given the small number of studies and limited long-term outcome data, it remains uncertain whether donor hypothermia improves long-term graft or patient survival.

Future studies should further analyse the impact of donor hypothermia in recipients of organs from extended criteria donors (ECD), as subgroup analysis based on donor criteria was limited in this review. Additionally, further studies are needed to assess the potential benefits of donor hypothermia in ECD donors, especially in the setting of machine perfusion, as this also remains an important question for future transplant practice.

Equity-related implications for research

Future work should evaluate the impact of donor hypothermia in low- and middle-income settings.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: 10.1002/14651858.CD015190.pub2.

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Analyses

Supplementary material 5 Data package

Supplementary material 6 Risk of bias assessment tool

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

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The following people conducted the editorial process for this article.

- Sign-off Editor (final editorial decision): David W Johnson, Department of Kidney and Transplant Services, Princess Alexandra Hospital, Brisbane, Australia;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Sam Hinsley, Central Editorial Service;
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- Copy Editor (copy editing and production): Narelle Willis, Central Production Service;
- Peer-reviewers (provided comments and recommended an editorial decision): Chiara Lazzeri, Tuscany Regional Transplant Center (clinical/content review); Clare Miles, Evidence Production and Methods Directorate (methods review); Jo Platt, Central Editorial Information Specialist (search review).
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Contributions of authors

- Draft the protocol: TH, ST, ET, CW
- Study selection: TH, ST
- Extract data from studies: DA, ST, TH
- Enter data into RevMan: DA, ST, TH
- Carry out the analysis: DA, ST, TH
- Interpret the analysis: DA, ST, TH, ET, CW
- Draft the final review: DA, ST, TH, ET, CW
- Disagreement resolution: ST, ET, CW
- Update the review: DA, ST, TH, ET, CW

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- Dharesh Raj Amarnath: no relevant interests were disclosed
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Data, code and other materials

As part of the published Cochrane review, the following are made available for download for users of the Cochrane Library: full search strategies for each database; full citations of each unique Cochrane Database of Systematic Reviews report for all studies included, or excluded at the full-text screen, in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows. Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, Review Manager, using the inbuilt computation methods. The data package is available in Supplementary material 5.

History

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ADDITIONAL TABLES

Table 1. Overview of included studies and syntheses

Study ID (country)[a]	Type of intervention	Comparator	Number of donors randomised (hypothermia/normothermia)	Number of donors who had at least one kidney transplanted (hypothermia/normothermia)	Number of recipients	Organs assessed	Duration at target temperature	Outcomes assessed with available data
HYPOREME 2024 (France)	Donor hypothermia (target core body temperature of 34°C to 35°C)	Donor normothermia (target core body temperature of 36.5°C to 37.5°C)	183/182	142/156	254/277	Kidney	Mean (SD): 2.8 (± 3.8)	• DGF (kidney) • 1-year graft survival (kidney) • 1-year patient survival (kidney) • Number of transplanted organs (kidney) • Donor adverse effects
Patel 2024 (USA)	Donor hypothermia (target core body temperature of 34°C to 35°C)	Donor normothermia (target core body temperature of 36.5°C to 37.5°C)	257/260	236/245	460/474	Kidney	Not available	• DGF (kidney)
Malinoski 2023 (USA)	Donor hypothermia (target core body temperature of 34°C to 35°C)	Donor normothermia (target core body temperature of 36.5°C to 37.5°C)	488/332	395/270	479/510	Kidney Heart Lung Liver Pancreas	Mean (SD): 30.3 (± 17.0)	• DGF (kidney) • 1-year graft survival (kidney) • Number of transplanted organs (kidney, heart, liver, lung, pancreas) • Donor adverse effects
Niemann 2015 (USA)	Donor hypothermia (target core body temperature of 34°C to 35°C)	Donor normothermia (target core body temperature of 36.5°C to 37.5°C)	197/197	150/152	285/287	Kidney Heart Lung Liver Pancreas	Median (IQR): 16.9 (10.2 to 29.4)	• DGF (kidney) • 1-year graft survival (kidney, lung, liver) • Number of transplanted organs (kidney, heart, liver, lung, pancreas) • Donor adverse effects

[a]See Supplementary material 2 for the full characteristics of the included studies
DGF: delayed graft function; IQR: interquartile range; SD: standard deviation